

What's New in Spectrum Analysers

If you plan to upgrade your test lab, you would like to get the latest spectrum analyser available in the market. This article talks about the latest cutting-edge spectrum analysers that are out in the market, covering their pricing, new technology and improved functionality

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There are general-purpose spectrum analysers, and then there are specialised ones. What kind of a spectrum analyser do you need? The answer to this question basically depends on the maximum frequency range that you want to measure, after including the harmonics and intermodulation products of the wanted signals. Although you can get a lab-grade oscilloscope that can analyse low audio frequencies up to microwave, this is overkill if you are going to use it with audio equipment. On the other hand, if you are working on very high frequencies, you might need an analyser that has a resolution good enough to differentiate between the smaller frequencies.

Some new releases in this space are:

1. Tektronix MDO4000 mixed-domain oscilloscope, which is claimed to be the world's first oscilloscope with built-in spectrum analyser.

2. R&S FSW-K91ac signal and spectrum analyser, which supports the new IEEE 802.11ac WLAN standard.



N9344C handheld spectrum analyser

3. Anritsu MS2830A + OML Mxx-HWID, which has a frequency coverage of up to 325 GHz.

4. Agilent N9344C HSA—an MIL PRF 28800 Class 2 compliant handheld spectrum analyser.

5. Scientech DSA800 Series spectrum analyser, which comes with a wide-screen display and is targeted at hobbyists.

The innovative lot includes:

1. Aronia X-Series USB RF spectrum analyser. Featuring specifications and performance similar to the equivalent Spec-

tran handheld spectrum analysers, these RF spectrum analysers are controlled entirely via the USB interface using the real-time RF analysis software included with every unit.

2. Metageek WiSPY DBX—a pocket-sized tool designed for enterprise-level troubleshooting of WiFi environments. It has

an amplitude range of -6.5 dBm to -100 dBm and amplitude resolution of 0.5 dBm. The device features an RP-SMA antenna too.

Purchasing an instrument for spectrum analysis being a major investment, make sure that you buy one that not only meets your today's requirements but can also cope-up with the demands of tomorrow. With this in mind, selecting the ideal instrument should be a well-thought-out affair, with sufficient inputs from the people who are going to use it. Here are some pointers.

1 Working with the IEEE 802.11ac WLAN standard?

If your project requires you to work with the upcoming 802.11ac WLAN standard, go for an analyser that is capable of analysing these signals. Although 802.11ac is still being defined, companies that chair the WiFi alliance have already brought out their own solutions using this standard.

The new standard requires 256 quadrature amplitude modulation (QAM), up to eight MIMO data

Key Features of R&S FSW-K91ac

Features	Value
Wide frequency range	2 Hz – 26.5 GHz
Low phase noise	-137 dBc (1Hz) at 10kHz offset (1GHz carrier)
WCDMA ACLR measurements	-88dB dynamic range (with noise cancellation)
Wide bandwidth	Up to 160MHz analysis bandwidth
Touch screen	High-resolution touchscreen display (31cm)
Parallel applications	Multiple measurement applications can be run and displayed in parallel
Very low EVM	Less than -45 dB